MIL-E-1/1330E
9 July 1975
SUPERSEDING
MIL-E-1/1330D
29 November 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID, MICROWAVE

TYPE 7486

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, metal-ceramic
See figure 1
Mounting position: Any
Weight: 0.06 ounce (1.7 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg
Unit:	V	Vdc	Vdc	v	ohms	Meg
Maximum:	6.6	250	0	+50	---	0.01
Minimum:	6.0	---	-50	-50	---	---
TEST CONDITIONS:	6.3	150	0	---	82	---

ABSOLUTE RATINGS:

Parameter:	Ik	Ik	Ic	Pp	TE	Barometric pressure, reduced
Unit:	ma	mAdc	mAdc	W	°C	mmHg
Maximum:	40.0	11.0	2.2	1.0	250 (Note 1)	8
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(E)	Qualification inspection								
1002	Barometric pressure, reduced	4	Voltage = 300 Vac	---	---	---	---	---	---
1031	Sweep-frequency vibration	10	Ebb = 150 Vdc; Rp = 10,000 ohms; 10 G; F = 100 to 2,000 Hz	---	---	Ep	---	15	mVac
1031	Sweep-frequency vibration fatigue	11	Eb = Ec = 0; Ef = 6.3 V	---	---	---	---	---	---
	Sweep-frequency vibration fatigue test end points:								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	µAdc
1306	Transconductance(1)	-		---	---	Sm	6,500	---	µmhos
1301	Heater current	-		---	---	If	222	258	mA
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	222	258	mA
1336	Heater-cathode leakage	-		0.65	II	Ihk	---	20	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	4.5	11.0	mAdc
1306	Transconductance (1)	-	Eb = 100 Vdc; Rk = 0	0.65	II	Sm	8,000	---	µmhos
1231	Pulsing emission	2	Rk = 1.0 ohm	0.65	II	ik	90	---	ma
1201	Short and discontinuity detection	-		0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	-		---	---	R	100	---	Meg
1306	Transconductance (2)	-	Ef = 6.0 V; Eb = 100 Vdc; Rk = 0	---	---	ΔSm Ef	---	20	%
1261	Electrode voltage (grid)	-	Rk = 0; Ec/Ib = 100 µAdc	---	---	Ec	---	-4.5	Vdc
1316	Amplification factor	-		---	---	Mu	65	115	---
1266	Grid emission	3	Ef = 7.0 V; Ecc = -20 Vdc; Rg = 0.1 Meg	---	---	Ic	0	-2.0	µAdc
3210	Grid recovery	7	Ebb = 250 Vdc; Ec/Ib = 3.0 mAdc; Rp = 0.01 Meg	---	---	ΔIb ΔIb	--- ---	0.6 1.0	mAdc ma

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1331	Direct-interelectrode capacitance	-	DESC Drawing No. 67030	---	---	Cgp	1.06	1.36	pF
						Cin	1.20	2.10	pF
						Cout	0.005	0.015	pF
						Chk	0.80	1.40	pF
---	Envelope strain	5		---	---	---	---	---	---
(E) 1031	Low-frequency vibration	9	Ebb = 150 Vdc; Rp = 10,000 ohms; 15 G; F = 40 Hz	---	---	Ep	---	10	mVac
	Quality conformance inspection, part 3								
1506	Heater-cycling life	8	Ef = 7.0 V; Ehk = +70 V; Rk = 0; Ec = Eb = 0	---	---	---	---	---	---
---	Heater-cycling life-test end point:								
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	μAdc
1516	Stability life	6	Ebb = 150 Vdc; Rk = 0; Eg = 7.7 Vac; Ehk = -70 Vdc; Rg = 3,300 ohms; Rb = 4,700 ohms; TA = 20°C	2.5	Code F	---	---	---	---
---	Stability life-test end point: (2 and 20 hours)								
---	Change in transconductance (1) of individual tubes	-		---	---	ΔSmt	---	15	%
1501	Intermittent life	-	Group A						
---	Operation (1)	-	Stability life-test conditions; t = 1,000 hours	---	---	---	---	---	---
---	Operation (2)	-	Ebb = 150 Vdc; Ec = -10 Vdc; ec = +8 v (driver side of Rg); Rg/g = 50 ohms; Rb = 60 ohms; tp = 10 ±1 μs; Du = 0.01 ±0.1%; Ehk = -70 Vdc; t = 1,000 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
	Intermittent life-test end points:								
---	Operation (1)	-							
1301	Heater current	-		---	---	If	222	264	mA
1306	Transconductance (1)	-		---	---	Sm	6,500	---	µmhos
1306	Transconductance (2)	-		---	---	ΔSm Ef	---	25	%
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
--	Operation (2)	-							
1301	Heater current	-		---	---	If	222	264	mA
1231	Pulsing emission	-		---	---	Ik	60	---	ma
1336	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
(E) 1041	Shock	12	Hammer angle = 30° Ehk = 50 Vdc	---	---	---	---	---	---
	Shock-test end points								
1031	Low-frequency vibration	-		---	---	Ep	---	15	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)	-		---	---	Sm	6,500	---	µmhos
1301	Heater current	-		---	---	If	222	258	mA

NOTES:

1. Operation below the rated maximum temperature is recommended for applications requiring the longest possible tube life.
2. Ec = -10 Vdc. Grid is driven with the pulse circuit as follows: ec = +7V; prr = 1,000; tp = 10 ±1µs; tr = 1µs (max); tf = 1µs maximum. Read peak cathode current.
3. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ecc	Eb	Rk	Rg
V	Vdc	Vdc	Ohms	Meg
7.0	0	150	82	0.1

4. The tube shall be tested as specified in method 1002 with the exception that the specified voltage shall be applied between the anode and grid.
5. Tubes shall be tested as specified in method 2126, except that they shall be first immersed in water at not more than 5°C for 15 seconds and immediately thereafter subjected to the standard temperature cycle specified in method 2126. Tubes having high heater current of 300 mA or more shall be rejected as evidence of air leaks.

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NOTES: - Continued

6. Eg = 7.7 Vac, 60 Hz, applied to grid through 100 μF coupling capacitor. Grid is connected to ground through 3,300 ohms.

7. Grid recovery. Test shall be performed as follows:

   (a) With TUT operating under specified conditions, adjust Ec for Ib = 3 mAdc.
   (b) Apply 5 volt pulse, 20 μs duration, 60 pps to the grid.
   (c) With application of pulse, measure undershoot (Δib) and change in average current (ΔIb).

8. The heater-cycling life-test sample shall consist of 15 tubes and no tube failure shall be permitted. In the event of rejection of the first sample due to failure of 1 tube, a second sample of 15 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed one.

9. Low-frequency vibration. Test shall be performed as follows: The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the heater contacts.

10. Sweep-frequency vibration. Test shall be performed as follows:

    (a) The frequency shall be increased from 100 to 2,000 Hz with approximately logarithmic progression in 3 ±1 minutes. The return sweep (2,000 to 100 Hz) is not required.

    (b) The tube shall be vibrated with simple harmonic motion in each of two planes; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and perpendicular to a line through the heater contacts. At all frequencies from 100 to 2,000 Hz, the total harmonic distortion of the acceleration waveform shall be less than 5 percent.

    (c) The peak acceleration shall be maintained at 10 ±1.0 G throughout the test.

    (d) The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of the vibration shall be measured with a suitable device having a response to the RMS value of the voltage to within ±0.5 dB of the response at 400 Hz for the frequency range of 100 to 3,000 Hz, and having a bandpass filter with an attenuation rate of 24 dB per octave below the low-frequency cut-off point of 50 Hz and above the high-frequency cut-off point of 5,000 Hz. The meter shall have a dynamic response characteristic equivalent to or faster than a VU meter (operated in accordance with Standard ASA No. C 16.5-1954).

11. Sweep-frequency vibration fatigue. Test shall be performed as specified in method 1031, except that the tubes shall be vibrated for a total of 6 hours, that is, 3 hours in each of two directions; first, parallel to the cylindrical axis; second, perpendicular to the cylindrical axis and parallel to a line through the heater contacts.

(E) 12. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

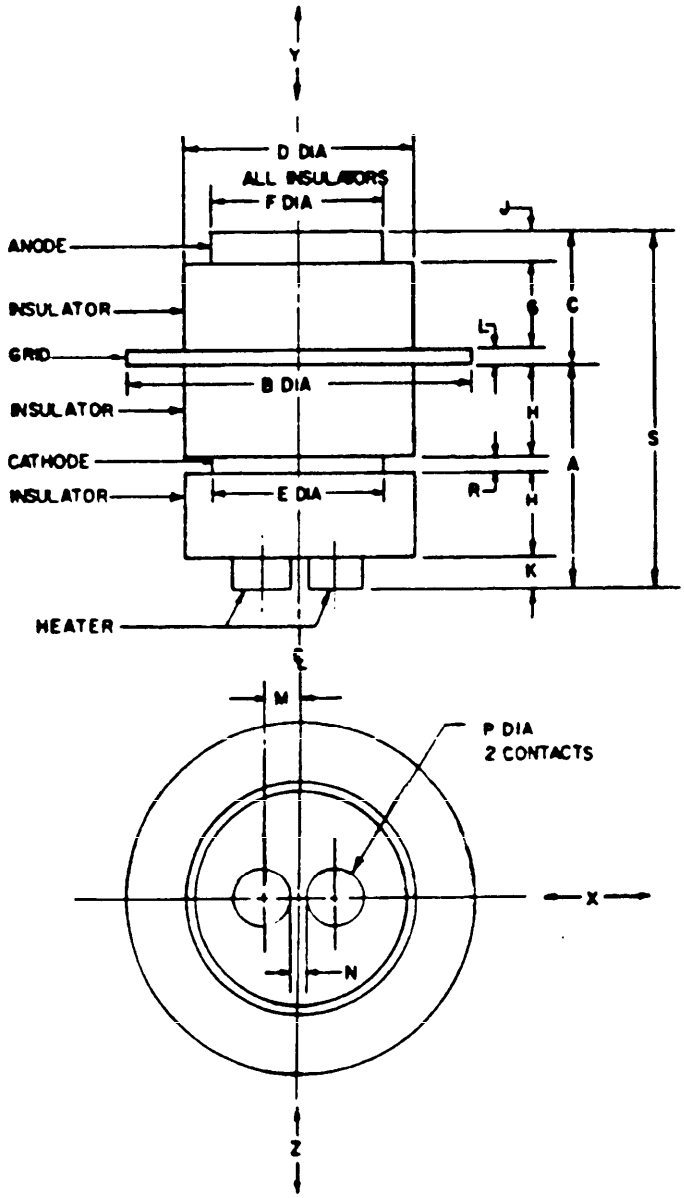
User activities:
Army - ME, MC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	0.268 (6.81)	0.292 (7.42)
B	0.476(12.09)	0.484(12.29)
C	0.156 (3.96)	0.174 (4.42)
D		0.335 (8.51)
E	0.281 (7.14)	0.289 (7.34)
F	0.271 (6.88)	0.279 (7.09)
G	0.095 (2.41)	0.105 (2.67)
H	0.095 (2.41)	0.105 (2.67)
J	0.034 (0.86)	0.046 (1.17)
K	0.047 (1.19)	0.063 (1.60)
L	0.024 (0.61)	0.030 (0.76)
M	0.055 (1.40)	0.081 (2.06)
N	0.030 (0.76)	
P	0.086 (2.18)	0.094 (2.39)
R	0.022 (0.56)	0.028 (0.71)
S	0.430(10.92)	0.460(11.68)

NOTES:
a. Eccentricity of anode, grid, and cathode with respect to centerline shall be 0.005 inch (0.127 mm) maximum.
b. Eccentricity of insulators with respect to centerline shall be 0.010 inch (0.25 mm) maximum.
c. Centerline of grid shall be reference line for horizontal tolerances.
d. Bottom surface of grid shall be reference plane for vertical tolerances.

FIGURE 1. Outline drawing of electron tube type 7486.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No.

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involve use of the specification in procurement of products for ultimate use by the Department of Defense. Th is provided for obtaining information on the use of this specification which will insure that suitable p can be procured with a minimum amount of delay and at the least cost. Comments and the return of th will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. C and suggestions submitted on this form do not constitute or imply authorization to waive any portion referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PR MENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additio attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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